Transient, Steady-State, and Extinction Angle (Beta) Analysis of a Half-Wave RL Rectifier

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Tools: Analytical Differential Calculus & MATLAB/Simulink

Executive Summary:
Investigation of magnetic energy storage, phase lag (Theta), and diode extinction angle (Beta) in an inductive half-wave rectifier (Vs = 230 Vrms, f = 100 Hz, R = 20 Ohms) under 20 mH, 50 mH, and 100 mH loads. Simulink scope waveforms are presented on Pages 3–7, followed by complete handwritten KVL differential equation derivations (I_total = I_ss + I_dyn) and analytical-to-Simulink verification on Pages 8–20.

Section 1: Physical Intuition — Inductor Behaviour & Water-Wheel Analogy

- water wheel = rating of inductor(H)
- initial resistance of wheel = reactance (XL)
- momentum of wheel after water turn off = (release of stored current.)
- speed of the wheel = peak current (A)

In DC circuits : inductor is exposed to a constant current so a magnetic field will be produced but it will be constant and don't change over time and thus act as a normal wire and do not oppose the incoming source current.

In AC circuits : inductor is exposed to alternating current in a coil and thus making a alternating magnetic flux and thus providing reactance (resistance / opposition) to source according to lenz law.

The given inductor ratings are (20mh,50mh,100mh)

inductance	Water wheel (H)	Initial Resistance of wheel (XL)	momentum of wheel after water turn off
20mh	Less weight	Low	Low
50mh	Medium weight	Medium	Medium
100mh	Heavy weight	high	high

20mh:

- The 20mh inductor offers very low initial reactance to the current provided by the source thus making the angle between the voltage and current very less (theta).
- So, the peak current in this inductor will be high as the resistance offered by the inductor (reactance) is very low. (The water can push the wheel fast, easily making it to go its maximum speed fast !).
- Once, the diode is reverse biased, the source will be cut-off and the energy stored in the inductors magnetic field will be released back to the circuit, thus making the diode to conduct for a short period of time even after its ideal 180 degree turn off period. It is calculated as (Beta) .
- Since it offers low momentum (less magnetic field current storage capacity), it releases the current and the diode conducts in the negative region but for a very short period of time.
- So, when the average voltage DC is calculated it will be high as the negative conduction period will be low.

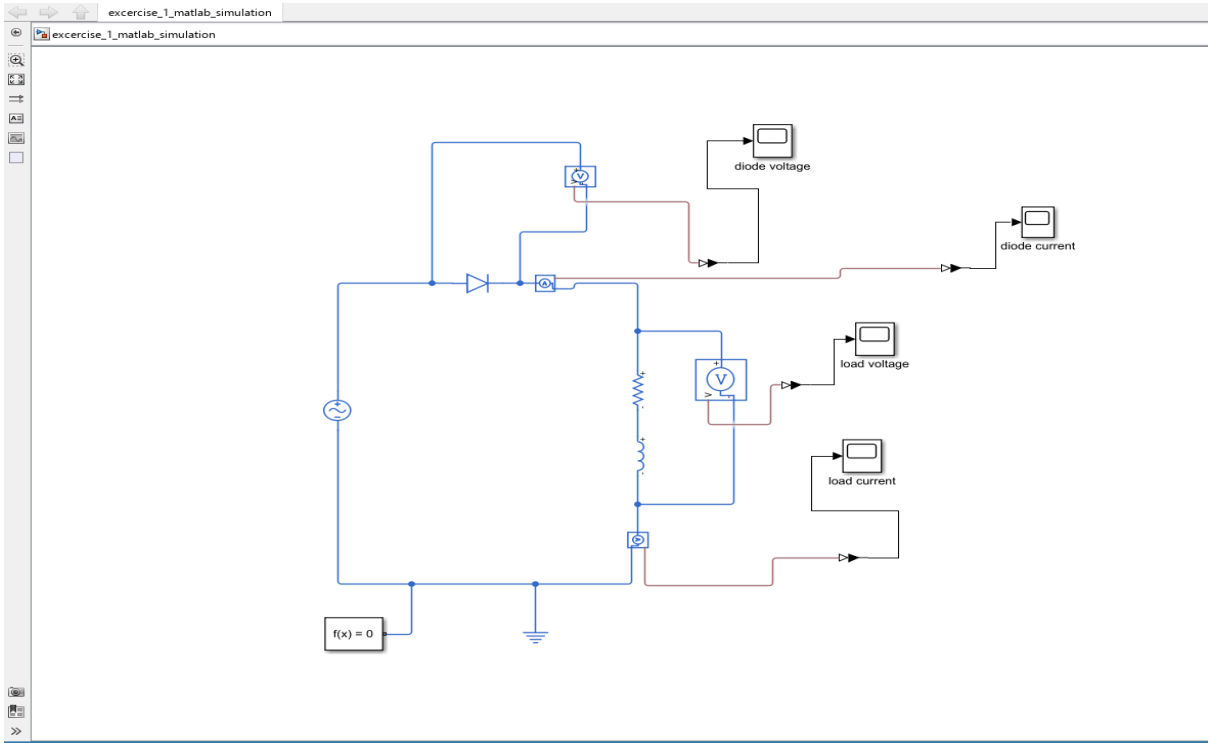
50mh:

- The 50mh inductor offers medium initial reactance compared to 20mh to the current provided by the source, thus making the angle between the voltage and current more compared to 20mh (theta).
- So, the peak current in this inductor will be lower compared to 20mh as the resistance offered by the inductor (reactance) is high compared to 20mh. (The water push the wheel at the same speed but wheel weight is heavy so takes a lot of time to achieve its maximum speed).
- Once, the diode is reverse biased, the source will be cut-off and the energy stored in the inductors magnetic field will be released back to the circuit, thus making the diode to conduct for a short period of time even after its ideal 180 degree turn off period. It is calculated as (Beta). Here the conduction angle at negative cycle will be high compared to 20mh.
- Since it offers more momentum than 20mh (more magnetic field current storage capacity), it releases the current and the diode conducts in the negative region but for a longer time compared to 20mh.
- So, when the average voltage DC is calculated it will be lesser than 20mh as the negative conduction period will be high compared to 20mh.

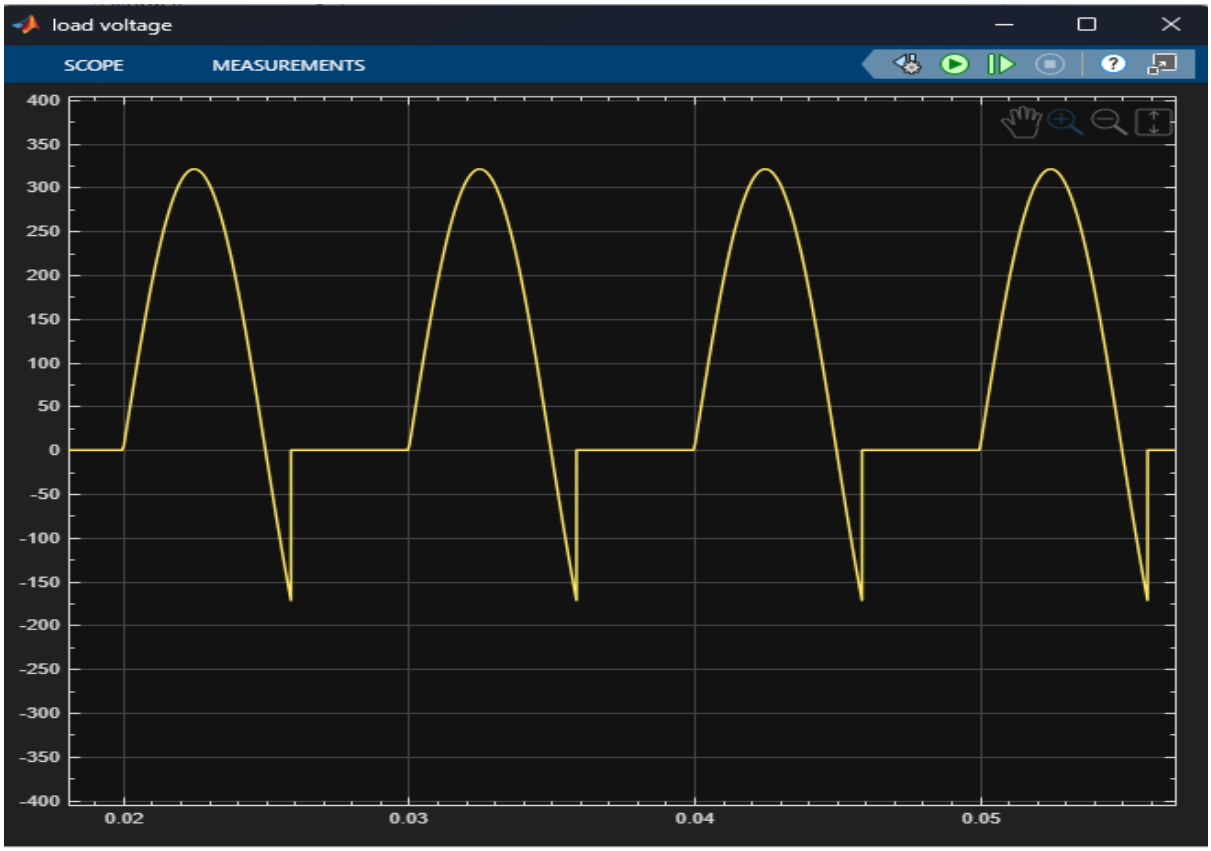
100mh:

- The 100mh inductor offers high initial reactance compared to 50mh to the current provided by the source, thus making the angle between the voltage and current higher compared to 50mh (theta).
- So, the peak current in this inductor will be lower compared to 50mh as the resistance offered by the inductor (reactance) is high compared to 50mh. (The water push the wheel at the same speed but wheel weight is heavy so takes a lot of time to achieve its maximum speed).
- Once, the diode is reverse biased, the source will be cut-off and the energy stored in the inductors magnetic field will be released back to the circuit, thus making the diode to conduct for a short period of time even after its ideal 180 degree turn off period. It is calculated as (Beta). Here the conduction angle at negative cycle will be high compared to 50mh.
- Since it offers more momentum than 50mh (more magnetic field current storage capacity), it releases the current and the diode conducts in the negative region for a longer time compared to 50mh.
- So, when the average voltage DC is calculated it will be lesser than 50mh as the negative conduction period will be high compared to 50mh.

Circuit diagram :



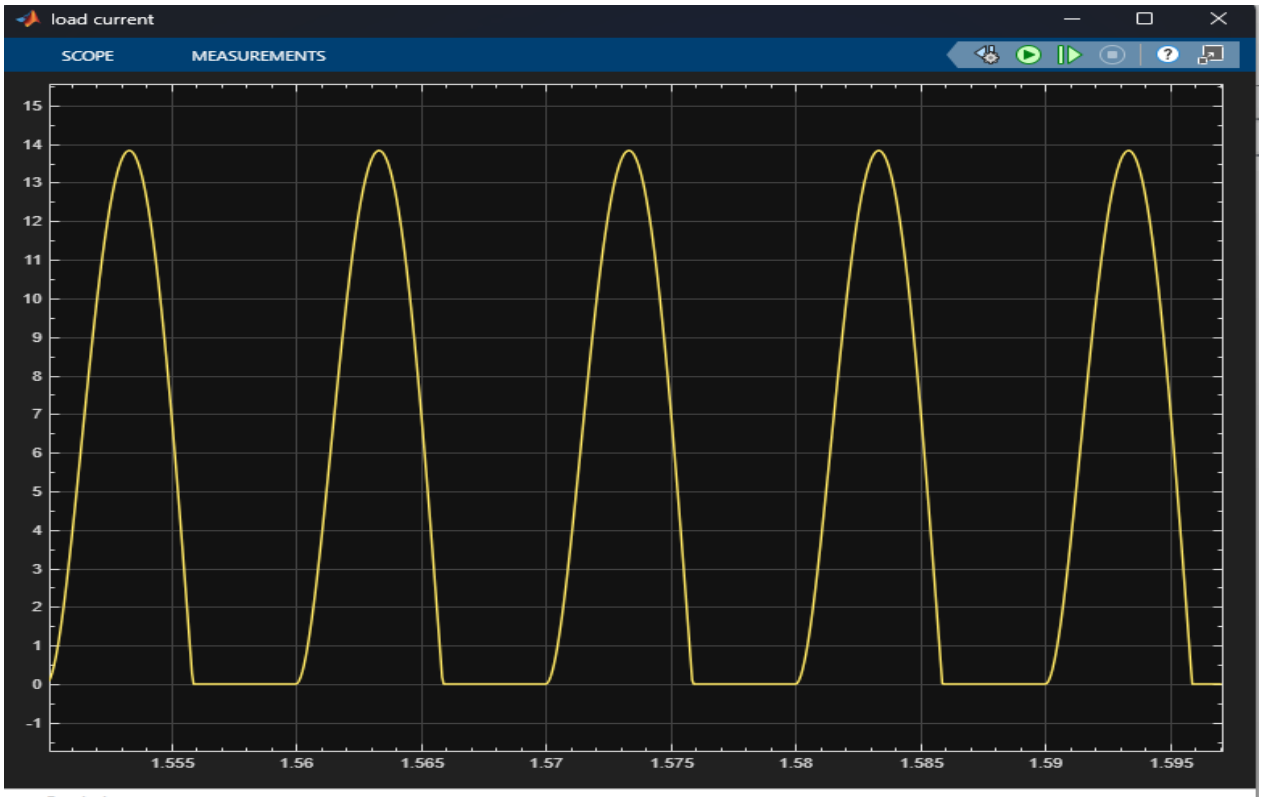
20mh load voltage:



Statistics

PS-Simulink C...	Max	Min	Peak to Peak	Mean	RMS
Time (ms)	22.4909	45.8809			
Value	321.0374290982129	-171.57762681397...	492.6150559121859	95.9108712324161	165.13092662178497

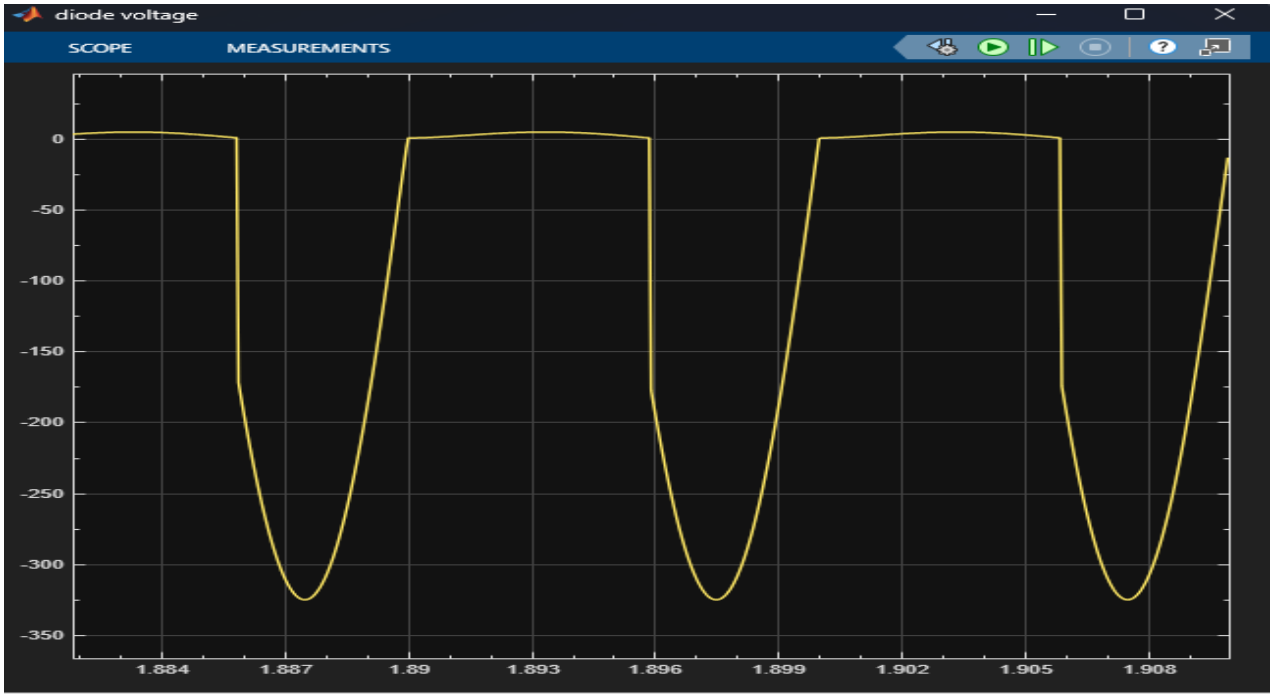
20mh load current:



Statistics

PS-Simulink Conv...	Max	Min	Peak to Peak	Mean	RMS
Time (s)	1.5733	1.5675			
Value	13.832369506623886	-0.00000325269053804...	13.832372759314424	4.962886329934818	7.2846328639253795

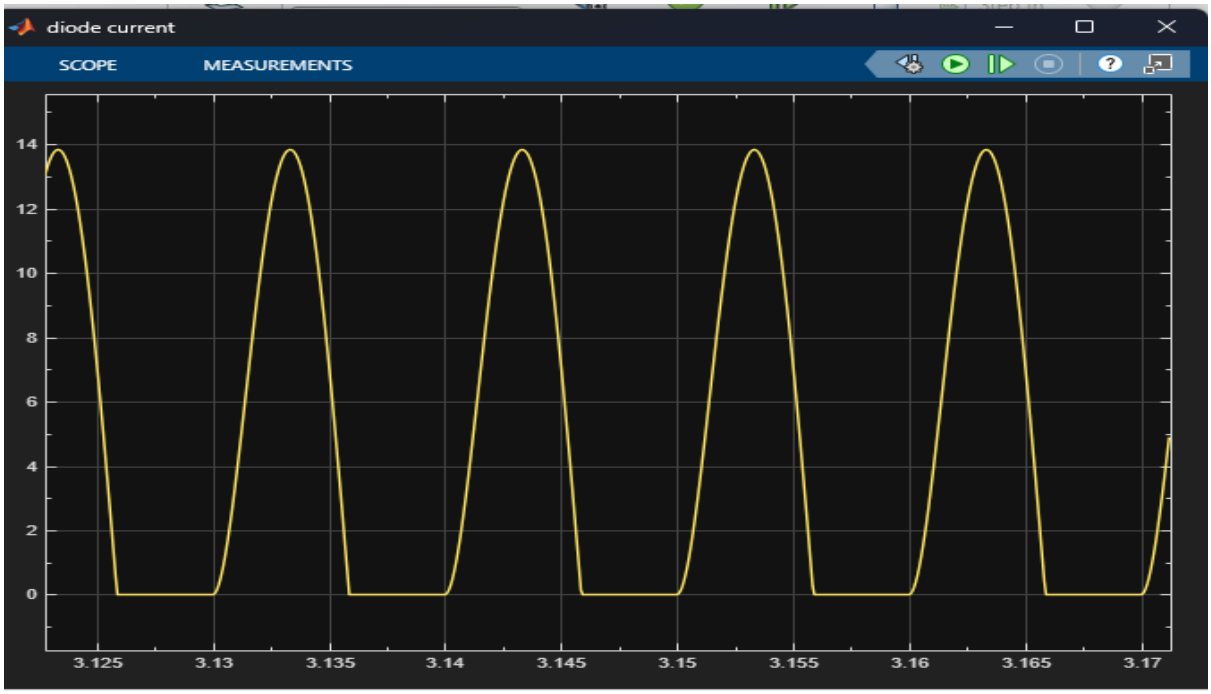
20mh diode voltage:



Statistics

PS-Simulink C...	Max	Min	Peak to Peak	Mean	RMS
Time (s)	1.9033	1.8875			
Value	4.749710850187087	-325.26905381128...	330.0187646614768	-100.80713644265...	165.030159456788...

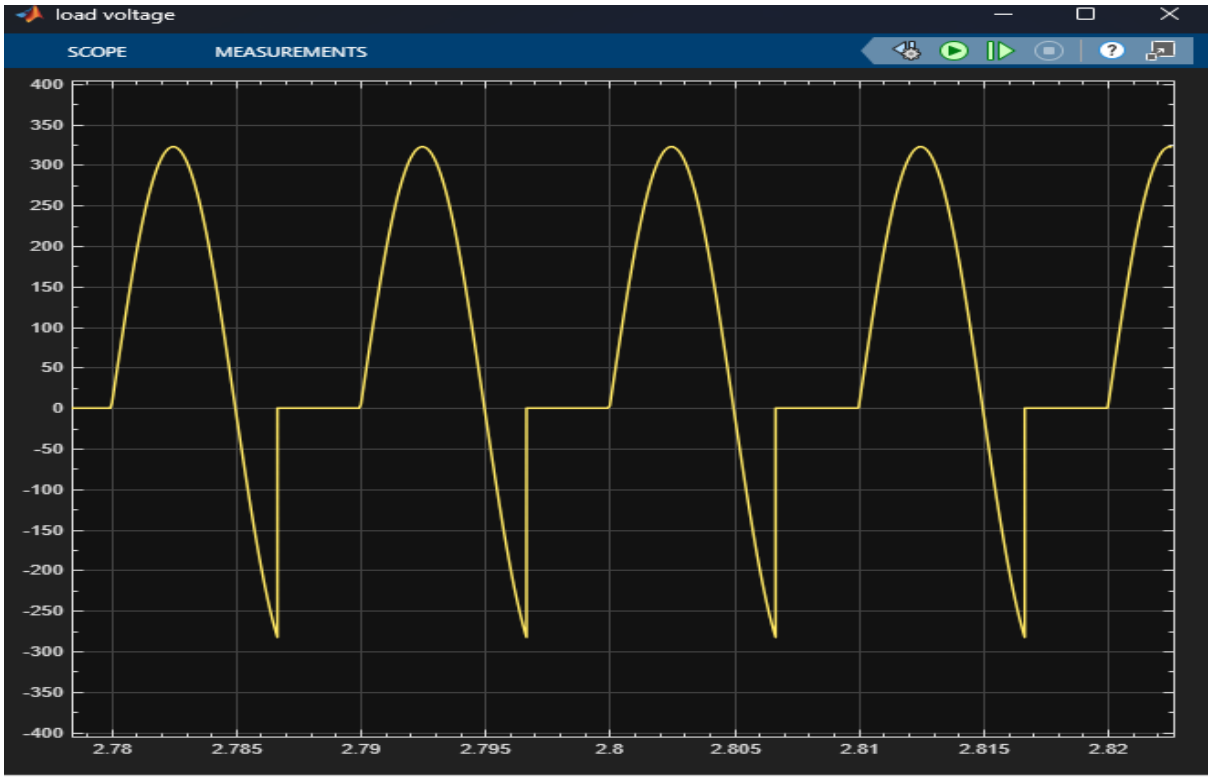
20mh diode current:



Statistics

PS-Simulink Co...	Max	Min	Peak to Peak	Mean	RMS
Time (s)	3.1633	3.1475			
Value	13.83236950662521	-0.00000325269053...	13.83237275931575	4.5097845565045755	6.955293020551888

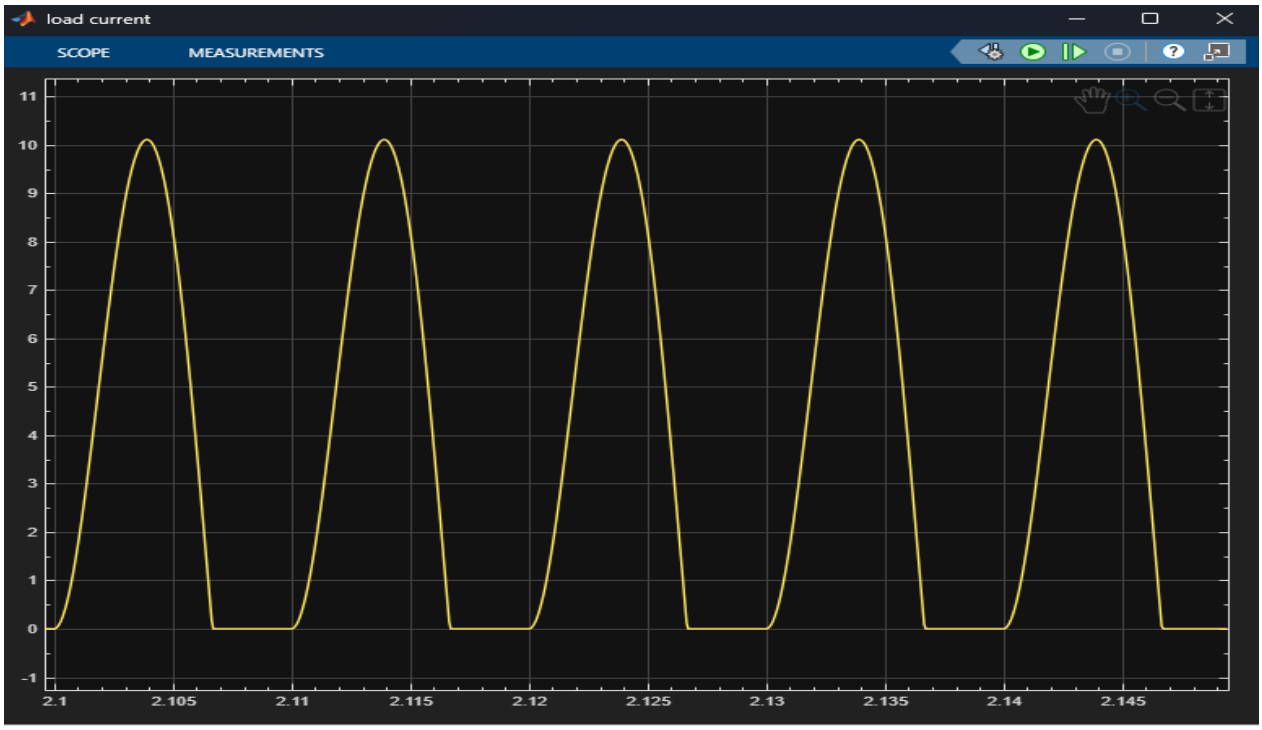
50 mh load voltage:



Statistics

PS-Simulink C...	Max	Min	Peak to Peak	Mean	RMS
Time	2.7825	2.7967			
Value	322.46912634710293	-282.7368486478693	605.2059749949722	80.2448672733155	177.00143085095152

50mh load current:

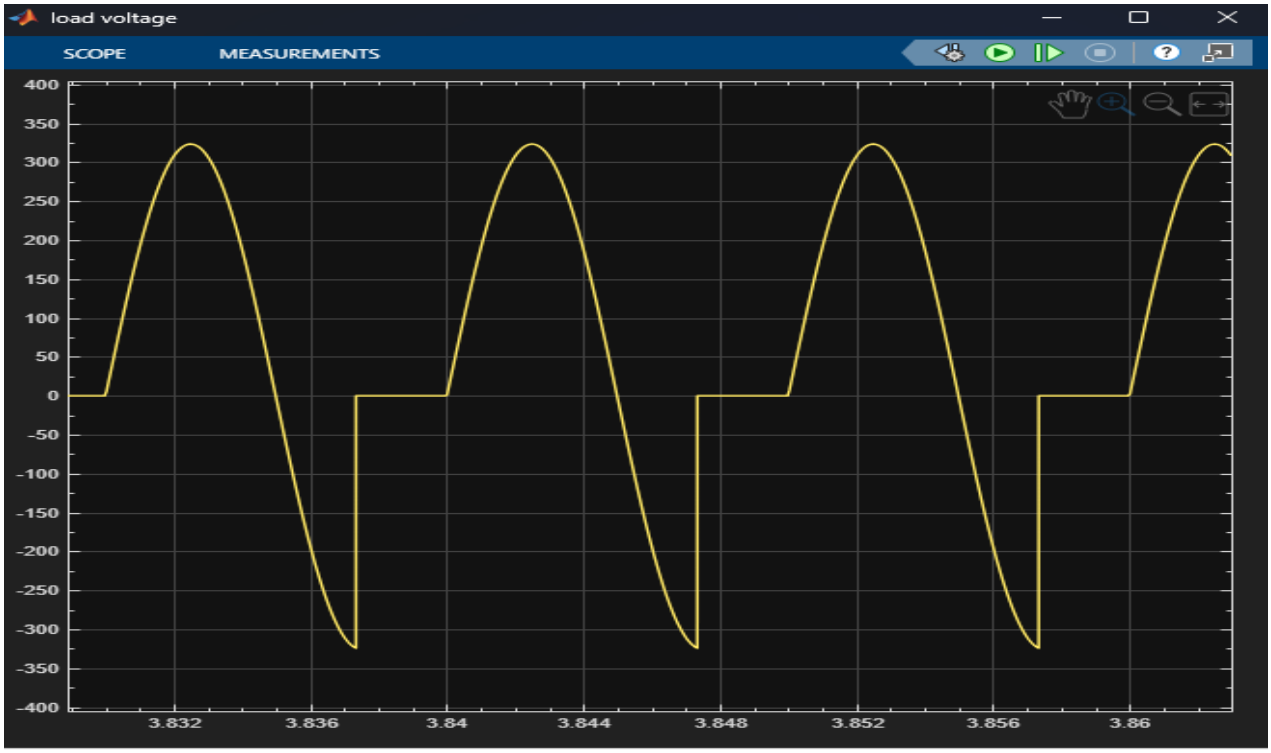


Statistics

PS-Simulink Conv...	Max	Min	Peak to Peak	Mean	RMS
Time (s)	2.1439	2.1175			
Value	10.112179055018071	-0.00000325269043280...	10.112182307708505	3.775967476024342	5.416215112673824

Ready Sample based T=5.0000

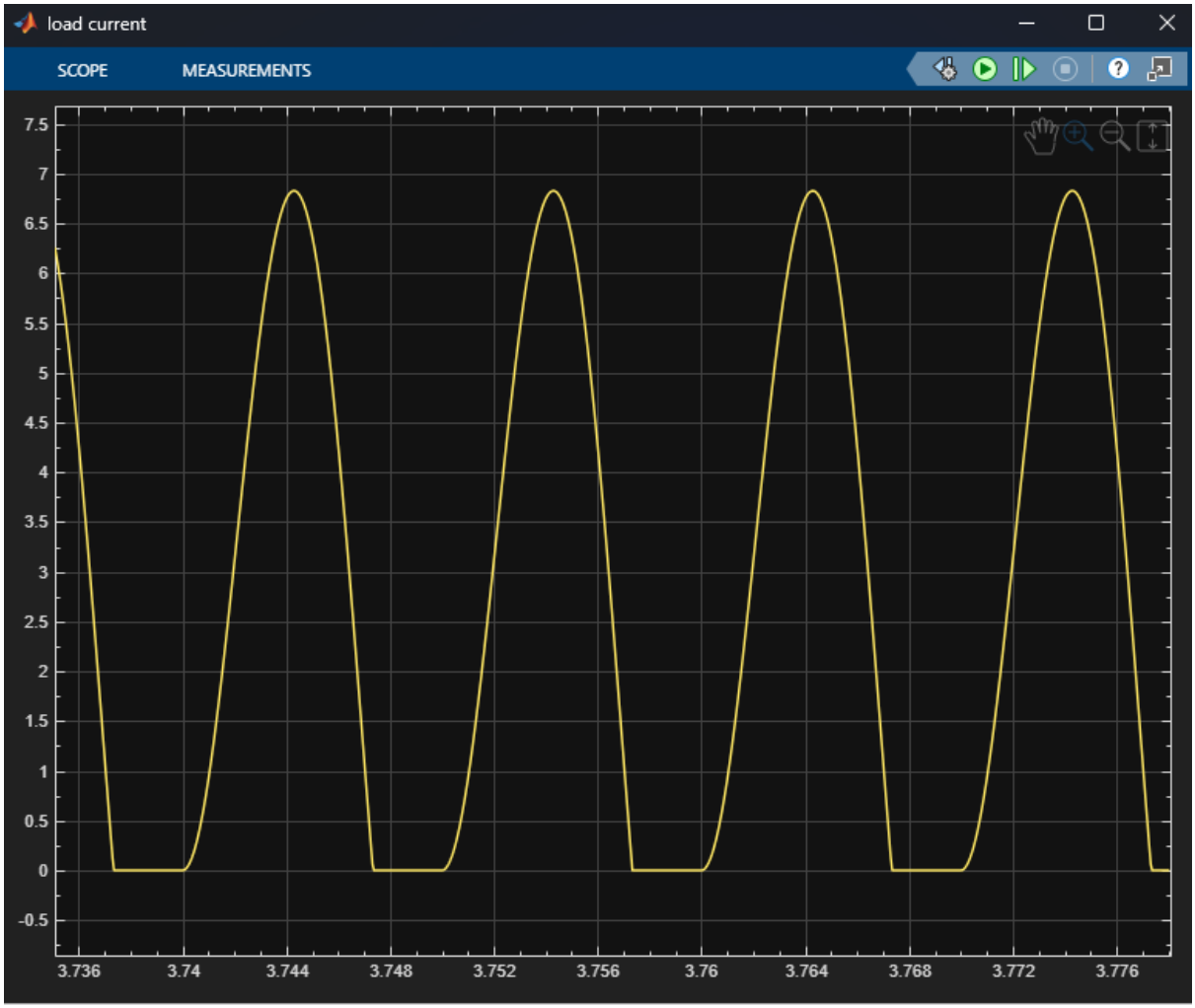
100mh load voltage:



Statistics

PS-Simulink C...	Max	Min	Peak to Peak	Mean	RMS
Time	3.8425	3.8373			
Value	323.37369674489486	-324.20878303453...	647.582479779426	67.64437957843161	195.7115008838602

100mh load current:



Statistics

PS-Simulink Conv...	Max	Min	Peak to Peak	Mean	RMS
Time (s)	3.7643	3.7675			
Value	6.827185859330324	-0.00000325269039378...	6.827189112020718	2.746383564364473	3.7707872672665834

Excercise (1): Given: $V_s = 230\,rms$, $F = 100Hz$, $R = 20\,\Omega$, $L = 20 \times 10^{-3}$.

① $V_s = V_m \sin(\omega t)$.

$\Rightarrow V_m = 230 \times \sqrt{2} = \boxed{325.269\,V = V_m}$

② $\omega = 2\pi F = 2 \times \pi \times 100 \Rightarrow \boxed{200\pi = \omega}$

③ $R = 20\,\Omega$, $X_L = \omega \times L = 200\pi \times 20 \times 10^{-3}$

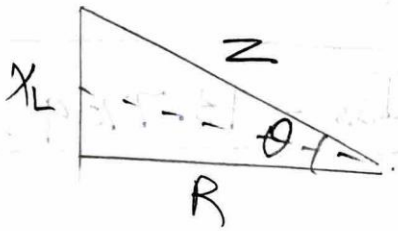
$\boxed{X_L = \omega L = 4\pi}$

④ $Z = \sqrt{R^2 + X_L^2} = \sqrt{(20)^2 + (4\pi)^2}$.

$\boxed{Z = 23.62\,\Omega}$

⑤ $\theta = \tan^{-1}\left(\frac{X_L}{R}\right) = \tan^{-1}\left(\frac{4\pi}{20}\right)$

$\boxed{\theta = 0.5609}$



$a^2 + b^2 = c^2$.

$Z = \sqrt{R^2 + X_L^2}$.

$\theta = \tan^{-1}\left(\frac{X_L}{R}\right)$

⑥ $V_{total} = V_m \sin(\omega t - \theta)$.

$\Rightarrow \boxed{325.269 \sin(200\pi t - 0.5609)}$

According to KVL law:

$$V_{Total} = V_{Resistor} + V_{inductor}.$$

$$\Rightarrow 325.269 \sin(200\pi t - 0.5609) = IR + L\frac{di}{dt}.$$

$I_{Total} = I_{ss} + I_{dyn}$ → To see the total current in the circuit at supply as well as when 0.

$$\Rightarrow 325.269 \sin(200\pi t - 0.5609) = I_{ss} R + L\frac{di_{ss}}{dt}.$$

$$Z = \sqrt{R^2 + X_L^2}.$$

$$\Rightarrow 325.269 \sin(200\pi t - 0.5609) = I_{ss} Z.$$

$$\Rightarrow I_{ss} = \frac{325.269 \sin(200\pi t - 0.5609)}{Z = 23.62}.$$

$$\boxed{I_{ss} = 13.7709 \sin(200\pi t - 0.5609)}$$

I_{dy} = V_{Total} is removed i.e, The Source is disconnected so the voltage in the circuit will be 0. But the inductor will discharge current.

$$0 = I_{dyn} R + L \frac{dI_{dyn}}{dt}.$$

$$-I_{dyn} R = L \frac{dI_{dyn}}{dt}.$$

$$\Rightarrow -\frac{I_{dyn} R}{L} = \frac{dI_{dyn}}{dt}.$$

$$-\frac{dt\, R}{L} = \frac{d\, I_{dyn}}{I_{dyn}}$$

$$\int -\frac{R}{L} dt = \int \frac{dI_{dyn}}{I_{dyn}}.$$

$$-R/L \int dt = \int \frac{1}{I_{dyn}} dI_{dyn}.$$

$\int c = cx + C$; $\int dt = t$; $\int \frac{1}{x} = \ln(x)$ or $\log_e(x)$.

$$\Rightarrow -\frac{R}{L} t + C = \ln(x) + C \qquad [x = I_{dyn}.]$$

$$\Rightarrow \ln(x) = -R/L\, t + C \qquad [\text{both } C \text{ are Considered as one}].$$

$$\Rightarrow x = e^{-R/L\, t + C}. \qquad x^a \times x^b = x^{a+b}.$$

$$\Rightarrow I_{dyn} = e^{-R/L\, t + C}.$$

$$\Rightarrow \boxed{I_{dyn} = e^{-R/L\, t} * e^{C}}$$

$$\boxed{I_{Total} = 13.7709 \sin(200\pi t - 0.5609) + e^{-R/L\, t} \times e^{C}.}$$

At initial Conditions, The Current will be zero and the time will be 0, The exakt Micro/nano second The Switch is turned on. So, $t = 0$, $I = 0$.

$$0 = 13.7709 \sin(200\pi t - 0.5609) + e^{-R/L t} \times e^{c}$$

here, The total Current at the instant will be zero so, $I_{total} = 0$.

The initial time will be zero, $t = 0$.

$$0 = 13.7709 \sin(200\pi(0) - 0.5609) + e^{-R/L(0)} \times e^{c}$$

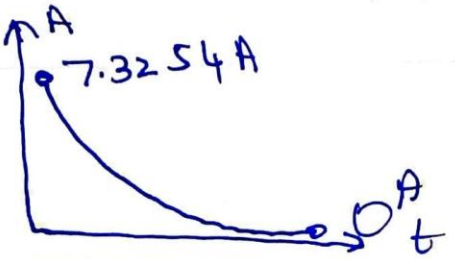
$$0 = 13.7709 \sin(0 - 0.5609) + 1 \times e^{c} \quad \sin(-\theta) = -\sin\theta$$

$$0 = 13.7709 \sin(-0.5609) + e^{c}$$

$$0 = -13.7709 \sin(0.5609) + e^{c}$$

$$13.7709 \sin(0.5609) = e^{c}$$

$$e^{c} = 7.3254\,A \rightarrow$$ The opposition provided by The Inductor initially So, as the pass =



Now, Substituting the current e^c in original equation.

$$I_{total} = 13.7709 \sin(200\pi t - 0.5609) + e^{-R/L\,t} \times 7.3254\,A.$$

*) Compared to time, the use of radians will be easy to calculate!

$\alpha = \omega t \quad t = \alpha/\omega \rightarrow$ substituting t value with alpha.

$$I_{total} = 13.7709 \sin(2 - 0.5609) + e^{-R/L \times \frac{2}{\omega}} \times 7.3254\,A.$$

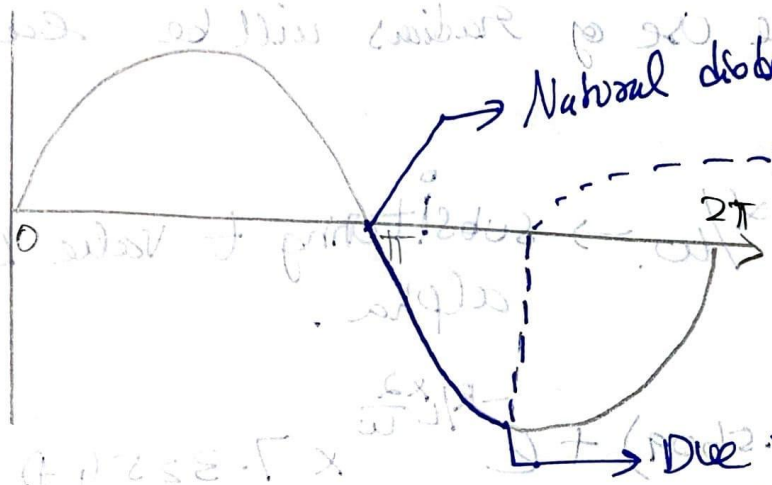
$$-\frac{R}{L} \times \frac{\alpha}{\omega} \Rightarrow \frac{-R}{\omega L} \times 2 \Rightarrow \frac{-20}{4\pi}\alpha \Rightarrow \boxed{-1.59154\,\alpha.}$$

$$\boxed{I(\alpha) = 13.7709 \sin(2 - 0.5609) + e^{-1.59154\,\alpha} \times 7.3254\,A.}$$

↳ Substitute the value of α in radians in the range 0 to β. to find the current value at any instant between 0 to β. (The total current in circuit at any time between 0 to β).

β = angle at which the diode turns off or reverse biased At this condition, the current will be definitely zero but inductor will release some current and make diode conductive for a short time

$$0 = 13.7709 \sin(\beta - 0.5609) + e^{-1.59154\beta} \times 7.3254$$



Natural diode turn off angle

----> assuming 270° as Conduction.

Due to inductive load, still conducts as inductor discharges.

$\beta = 3.70395.$

So, from zero to β the diode conducts and goes to 0 and cycle continues.

*) Average DC Voltage:

$$V_{DC(avg)} = \frac{1}{2\pi}\int_0^{\beta} \underbrace{\sin(\alpha)\, V_m}_{\text{normal voltage.}}\, d\alpha$$

① Range of α = 0 to β as diode only conducts that time.

② Voltage DC = $V_m \sin(\alpha)$ but, we take average so,

$$\frac{1}{2\pi}\int_0^{\beta} V_m \sin(\alpha)\, d\alpha.$$

$$I_{(avg)} = \frac{1}{2\pi}\int_0^{\beta} \sin(\alpha)\, V_m \, d\alpha \rightarrow (1)$$

$\sin \rightarrow -\cos \rightarrow -\sin \rightarrow \cos$

Integ.

$$V_{DC\,(avg)} = \frac{V_m}{2\pi}\int_0^{\beta} \sin(\alpha)\, d\alpha \rightarrow (2)$$

$\beta = 3.70395$

$$V_{DC\,(avg)} = \frac{V_m}{2\pi} \times \left[-\cos(\alpha)\right]_0^{\beta} \rightarrow (3).$$

$$V_{DC\,(avg)} = \frac{V_m}{2\pi} \times \left[\left[-\cos(\beta)\right] - \left[-\cos(0)\right]\right]$$

$$V_{DC\,(avg)} = \frac{325.269}{2\pi} \times \left[\left[-\cos(3.70395)\right] - \left[-\cos(0)\right]\right]$$

$V_{DC\,(avg)} = 95.56407\ V$ = This is also the Average load voltage

Root mean Square Voltage of the Dc.

RMS of V_{DC}.

$$V_{DC} = V_m \sin(\alpha).$$

Square

$$V_{DC} = (V_m \sin(\alpha))^2 \Rightarrow V_m^2 (\sin(\alpha))^2 \Rightarrow \sin^2\alpha.$$

$$\hookrightarrow V_m^2 \sin^2\alpha.$$

Mean

$$V_{DC} = \frac{1}{2\pi} \int_0^{\beta} (V_m \sin(\alpha))^2$$

Root

$$V_{DC\,(Rms)} = \sqrt{\frac{1}{2\pi} \int_0^{\beta} (V_m \sin(\alpha))^2}$$

$$\sin^2\theta = \frac{1-\cos(2\theta)}{2}$$

$$V_{DC\,(Rms)} = \sqrt{\frac{1}{2\pi} \int_0^{\beta} V_m^2 \sin^2\alpha}$$

$$V_{DC\,(Rms)} = V_m \sqrt{\frac{1}{2\pi} \int_0^{\beta} \sin^2\alpha}.$$

$$V_{DC\,(Rms)} = V_m \times \sqrt{\frac{1}{2\pi} \int_0^{\beta} \frac{1-\cos(2\alpha)}{2}}$$

$$V_{DC\,(Rms)} = V_m \times \sqrt{\frac{1}{2\pi} \int_0^{\beta} \frac{1}{2} - \frac{\cos(2\alpha)}{2}}$$

$$V_{(rms)} = V_m \times \sqrt{\frac{1}{2\pi}\left[\int_0^{\beta} \frac{1}{2} - \left[\frac{1}{2}\times\cos(2\alpha)\right]\right]}$$

$\int d = dx + c$

$+c$ is removed as we deal with Integral bounds.

V_{DC} (Rms) [Calculating the Integration part]

$$\int_0^{\beta} \frac{1}{2} - \int_0^{\beta} \frac{1}{2}\cos(2\alpha)$$

$\int_0^{\beta} \frac{1}{2}$ → A

$\int_0^{\beta} \frac{1}{2}\cos(2\alpha)$ → B.

$$A = \int_0^{\beta} \frac{1}{2} \Rightarrow \frac{1}{2}\alpha.$$

$$B = \int_0^{\beta} \frac{1}{2}\cos(2\alpha)$$

$$B = \frac{1}{2}\int_0^{\beta} \cos(2\alpha)$$

$\sin \rightarrow -\cos \rightarrow -\sin \rightarrow \cos$

α ← Integ.

$$\cos(kx) = \frac{1}{k}\sin(kx) + c.$$

$$B = \frac{1}{2} \times \left[\frac{1}{2}\sin(2\alpha)\right]$$

$$B = \frac{\sin(2\alpha)}{4}$$

$$\Rightarrow \left[\frac{1}{2}\alpha\right]_0^{\beta} - \left[\frac{\sin(2\alpha)}{4}\right]_0^{\beta}$$

$$\left[\frac{1}{2}(\beta) - \frac{1}{2}(0)\right] - \left[\frac{\sin(2\beta)}{4} - \frac{\sin(2(0))}{4}\right] \quad \beta = 3.705$$

$\Rightarrow$ Integration part $= 1.62643$

$V_m = 325.269\,V.$

$$V_{DC\,(RMS)} = V_m \times \sqrt{\frac{1}{2\pi} \times [1.62643]}$$

$$V_{DC\,(RMS)} = 325.269 \times \sqrt{0.25885\,4.}$$

$$\boxed{V_{DC\,(RMS)} = 165.4893\,V.}$$

Peak load voltage $= V_m$ as Diode will be in Conduction.

$$\boxed{V_{peak} = 325.27\,V}$$

Average value of AC is DC and in DC Inductor is normal so; $\boxed{I = V/R = \frac{95.56V}{20} \Rightarrow 4.77A}$

3.7 Load V = 325·269 × sin(α) → [0 to β]

Load I = 13.7709 sin(α − 0.5609) + $e^{-1.5\theta/1.5\alpha}$ × 7.325A → [0 to β]

Diode Voltage and Current : [across Diode].

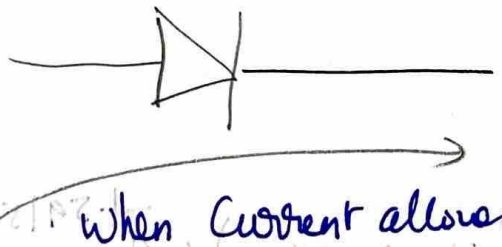
Current : 0 to β : $0 \le \alpha \le \beta$.

$$I(\alpha) = 13.77 \sin(\alpha - 0.561) + 7.33 \times e^{-1.5\alpha}$$

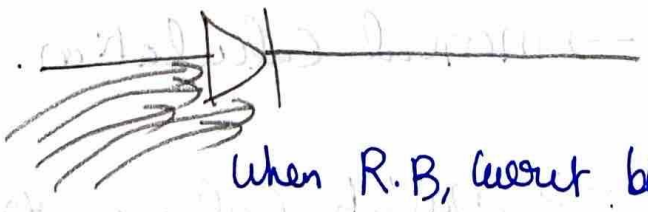
Current : after beta = 0 Then agn Cycle Repeats.

Voltage : $V_m \sin(\alpha)$. when Blocking.

0 when allowed $0 \le \alpha \le \beta$. } Repeats.



when Current allowed, F·B, no Voltage across diode. → V = 0



when R.B, Current blocked, Entire Voltage applied across Diode, Mainly the Voltage-ke across it

↳ $V = V_m \sin(\alpha)$.

Matlab and paper Analysis:

① Matlab: at First cycle of frequency, The Diode peak occures at 3.3443 sec and The Current value is 13.8240 A.

② Analysis in Made equation:

$$\left[I_2 = 13.7709 \sin(\alpha - 0.5609) + e^{-1.59154\alpha} \times 7.3254\ A.\right]$$

Matlab time to Radius.

$\omega \times t$ $\quad \omega = 2\pi F \times t$ (of first cycle)

$$\Rightarrow 2 \times \pi \times 100 \times 3.3443 \times 10^{-3} = \alpha.$$

$$\boxed{\alpha = 2.10128.}$$

$$I_2 = 13.7709 \sin(2.10128 - 0.5609) + e^{-1.59154 \times 2.10128} \times 7.3254\ A.$$

$I_2 = 14.02\ A \rightarrow$ Manual calculation. Diode

$I_{matlab} = 13.8240\ A \rightarrow$ Matlab Calculation of Diode

Same For load Current as Both will be same.

of load voltage = $V_{peak\ of\ sine} = V_m = 325.269$.

Matlab analysis of Peak load voltage = 321.0225

Load Voltage = $325.269 \times \sin(\alpha)$

$\alpha = 2\pi F \times 2.4734 \times 10^{-3}$ Sec.

$\alpha = 1.5540$

$\Rightarrow 325.269 \times \sin(1.5540) \Rightarrow 325.223V.$

$V_{DC\ RMS} = 164.694 \rightarrow$ Matlab.

$V_{DC\ RMS} = 165.48 \rightarrow$ Maths.

$V_{DC\ Mean\ (Avg)} \rightarrow 92.960V. \rightarrow$ Matlab.

$V_{DC\ Mean\ (Maths)} \rightarrow 95.56V \rightarrow$ Maths